8.5 MentorPi Servo Deviation Adjustment

1. Preface

If you purchased an Ackermann chassis or a monocular camera version, the system includes several PWM servos that serve as the core components for steering control. Due to assembly tolerances, you may encounter slight deviations—such as the vehicle veering slightly to one side—even after the servos are powered on and set to their neutral positions.

This document provides step-by-step guidance to help you manually adjust the servo alignment and correct these deviations.

2. Operation Steps

1) Start the robot and access the robot system using the remote control software VNC according to the instructions provided in '8. Remote Tool Installation and Container Access/ 8.1 VNC Installation and Connection'.

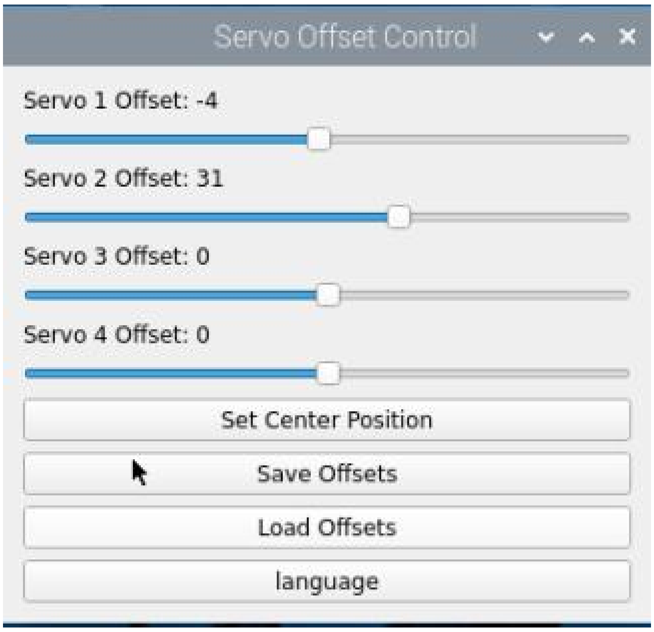
2) Execute the command below to disable the app auto-start service.

```
~/.stop_ros.sh
```

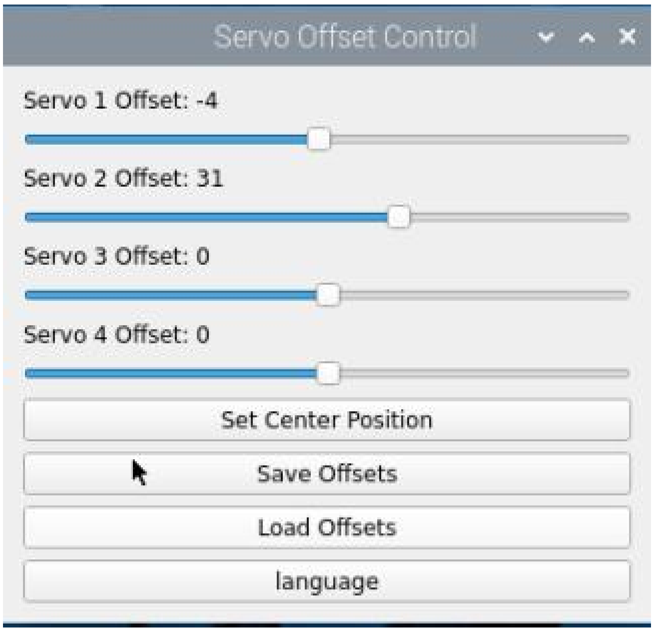
3) Click-on  and input the following command to run the deviation adjustment script.

```
./Servo.sh
```

4) After executing the command, a pop-up window will appear as shown below.



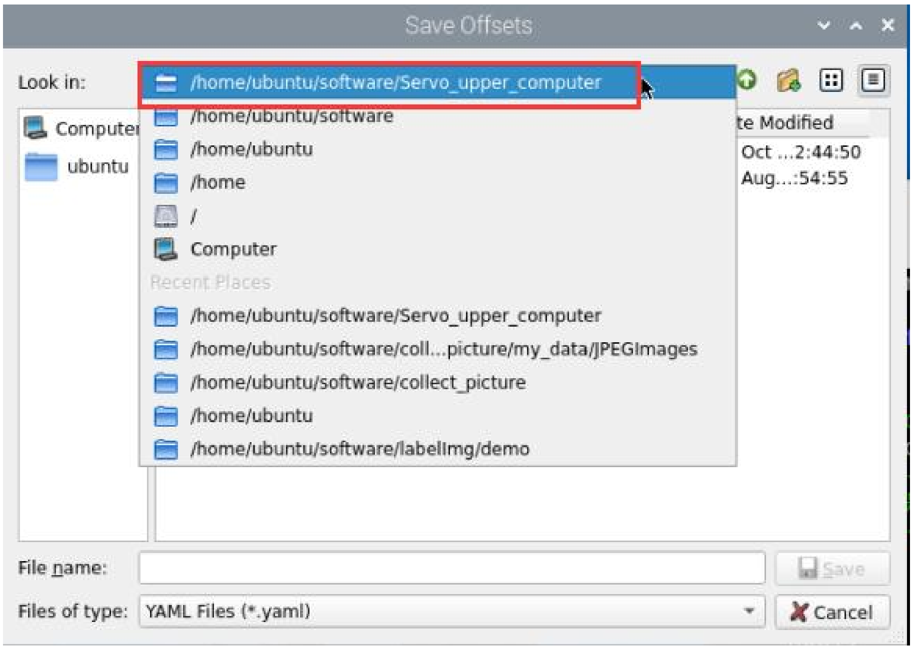
5) Drag the four sliders shown below to adjust the corresponding servo offsets. The servos will rotate in real time during the adjustment, allowing you to observe their movement and fine-tune the deviation accordingly.



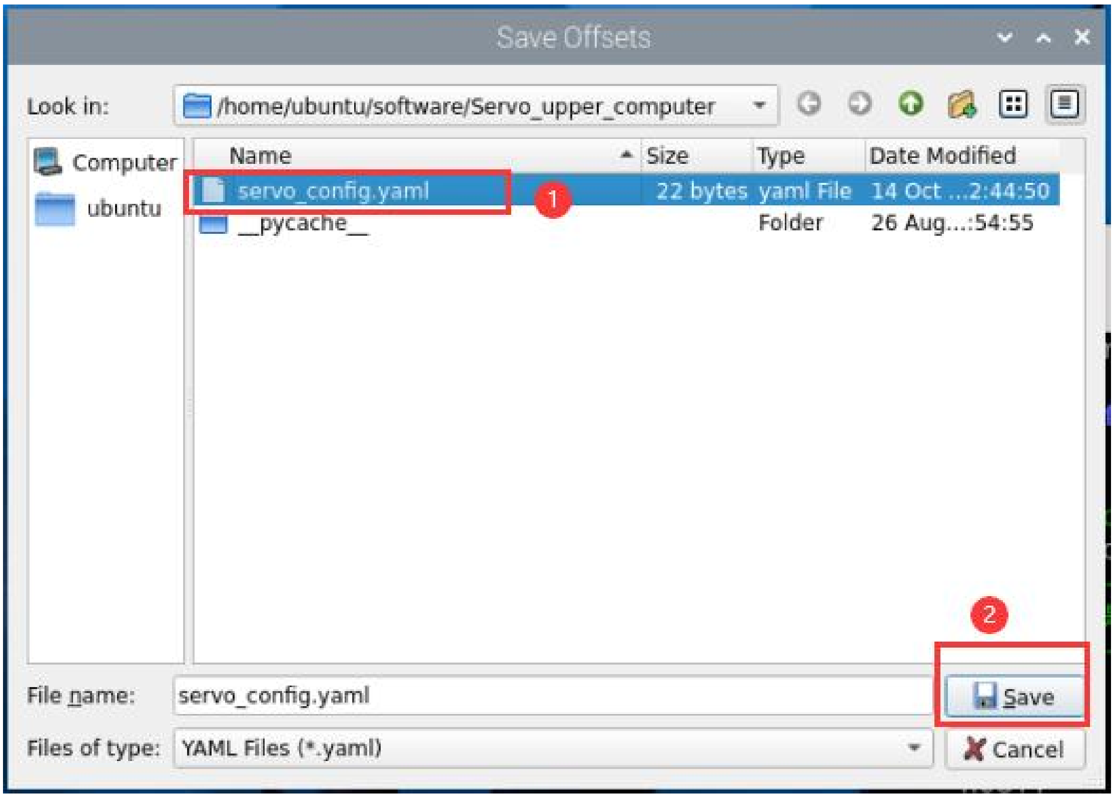
6) If you're not satisfied with the adjustment results, click the 'Set Center Position' button to reset all servo offsets to their default center positions and readjust as needed.

7) After adjusting the servos to the desired positions, click the 'Save Offsets' button to save the calibration settings.

Below is the storage path.



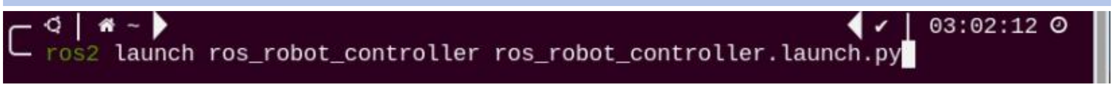
The following file will be saved:



After completing this step, the offset values will be stored. In all future applications involving the servos, this offset file will be automatically loaded to compensate for any servo deviations.

8) You may also enter the following command and press Enter to launch the chassis control node and verify that the offset values have been applied correctly.

```
ros2 launch ros_robot_controller ros_robot_controller.launch.py
```



It can be confirmed that the offset values have been successfully saved to the file and can be correctly loaded in other applications.

```
❯ ros2 launch ros_robot_controller ros_robot_controller.launch.py
[INFO] [launch]: All log files can be found below /home/ubuntu/.ros/log/2024-10-14-13-43-35-574879-raspberrypi-70165
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [ros_robot_controller-1]: process started with pid [70166]
[ros_robot_controller-1] [INFO] [1728913416.797319250] [ros_robot_controller]: 已设置舵机 1 的偏移量为 -4
[ros_robot_controller-1] [INFO] [1728913416.798871880] [ros_robot_controller]: 已设置舵机 2 的偏移量为 31
[ros_robot_controller-1] [INFO] [1728913416.799888139] [ros_robot_controller]: 已设置舵机 3 的偏移量为 0
[ros_robot_controller-1] [INFO] [1728913416.800728954] [ros_robot_controller]: 已设置舵机 4 的偏移量为 0
[ros_robot_controller-1] [INFO] [1728913416.804926973] [ros_robot_controller]: start
```